

# Problem Set - 11

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# 1 Relativity

## 1.1 Photon Box

An empty box of total mass  $M$  with perfectly reflecting walls is at rest in the lab frame. Then electromagnetic standing waves are introduced along the  $x$ -direction, consisting of  $N$  photons, each of frequency  $\nu_0$ .

(A Guide to Physics Problems, 2.4)

- a) State what the rest mass of the system (box + photons) will be when the photons are present.
- b) Show that this answer can be obtained by considering the momentum and/or energy of the box-plus-photon system in any inertial frame moving along the  $x$ -axis.

## 2 Modern Physics

### 2.1 Rotating with Light

A light source is emitting light with intensity  $I$  and wavelength  $\lambda$  on a sphere with angular momentum  $J$  parallel to its axis. If the light source is on for time  $\tau$ , what is the final angular velocity of the sphere?

(2024 4. Aşama)

## TD 7

### Forced Harmonic oscillator

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#### EXERCISE 0. WARM UP!

1. The original pendulum installed by Foucault in the Pantheon in Paris in 1851 aimed at demonstrating the rotation of the Earth. It consisted of a ball of lead and brass of mass  $m = 28$  kg and diameter  $d = 38$  cm attached to a steel wire of length  $\ell = 67$  m. The oscillating period of the pendulum was
  - $\sqrt{\frac{d}{g}}$
  - $\frac{m}{g}$
  - $2\pi\sqrt{\frac{\ell}{g}}$
  - smaller and smaller as time went by.
2. Give an estimate of the oscillating period of the Foucault pendulum.
3. On a guitar, the strings corresponding to the lowest tunes are:
  - thicker?
  - thinner?
4. A forced mass-and-spring system oscillates:
  - at the natural frequency  $f_0 = \omega_0/2\pi$  of the system.
  - at the frequency  $f = \omega/2\pi$  of the excitation.
5. The response of a forced oscillator is peaked around its natural frequency. If damping is increased, does the response peak sharpen or broaden?

#### EXERCISE 1. BUMPY ROAD AHEAD!

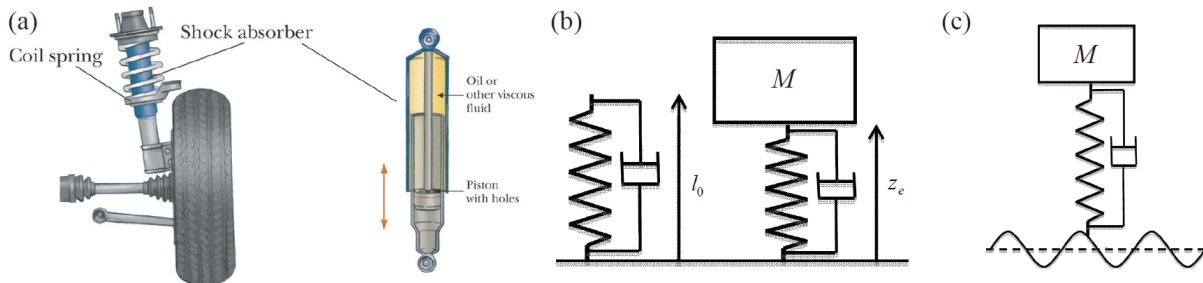


Figure 1: Model of a car suspension driving on a wavy road.

A car suspension is composed of a spring of stiffness  $k$  and rest length  $l_0$  attached to the car's chassis of mass  $M$ , and coupled to a dashpot (fluid damper) that opposes the movement of the car's chassis according to  $\vec{R} = -\alpha\vec{v}$ . As shown in Figure 1b, the system can be modeled by a block connected to the floor by a spring and a fluid damper.

1. The car is at rest on a flat road. List the forces acting on the chassis. Determine its equilibrium height  $z_e$  above the ground.
2. The car is now being driven over a bumpy road at constant velocity  $V$  (see Fig.1c). We assume that the bumps profile follows a cosine function  $\delta(x) = \delta_0 \cos(2\pi x/\lambda)$ , where  $\delta_0$  and  $\lambda$  are respectively the amplitude and wavelength of ground undulations. As the car moves at velocity  $V$ ,  $\delta$  varies in time according to:

$$\delta(t) = \delta_0 \cos(2\pi Vt/\lambda) = \delta_0 \cos(\omega t)$$

where  $\omega = 2\pi V/\lambda$ .

- (a) Write the differential equation of the movement satisfied by the vertical displacement  $z(t)$  of the chassis. Using the displacement of the car from its equilibrium height above the road  $u = z - z_e - \delta$ , rewrite the obtained differential equation as a function of this new variable.
- (b) Rewrite the equation under the form  $\ddot{u} + \frac{\omega_0}{Q}\dot{u} + \omega_0^2 u = G \cos(\omega t)$  and give the expression of the parameters  $\omega_0$ ,  $Q$  and  $G$  in terms of  $m$ ,  $\alpha$ ,  $k$ ,  $\omega$  and  $\delta_0$ .
- (c) The solution of this differential equation is of the form  $u(t) = u_H(t) + u_P(t)$  where  $u_H(t)$  is the general solution of the associated homogeneous equation and  $u_P(t)$  a particular solution of the full equation. What is the homogeneous equation associated to our differential equation?
- (d) Based on the behavior of the free damped oscillator you studied in the last tutorial, what is  $\lim_{t \rightarrow +\infty} u_H(t)$ ?
- (e) We consider the stationary state of the forced oscillation (at long times), where the transient response of the free oscillator has already decayed. Then  $u(t) = u_P(t)$ . What would be the form of this particular solution  $u_P(t)$ ?

### 3. Determination of the forced solution:

- (a) Show that  $u_P(t) = A \cos(\omega t + \Phi)$  can rewrite in complex notation as  $u_P(t) = \text{Re}(\tilde{u}e^{i\omega t})$  with  $\tilde{u}$  a complex number that you will express as a function of the constants  $A$  and  $\Phi$ . Then show that  $G \cos(\omega t)$  also writes as  $g = \text{Re}(Ge^{i\omega t})$ .
- (b) Using those complex notations, we are going to determine the long term response of the system to the forcing applied. Show that if  $U = \tilde{u}e^{i\omega t}$  is solution of the complex differential equation

$$\ddot{U} + \frac{\omega_0}{Q}\dot{U} + \omega_0^2 U = Ge^{i\omega t}, \quad (1)$$

then  $u_P = \text{Re}(U)$  would be solution of the differential equation of question 2(b).

- (c) Inject  $U = \tilde{u}e^{i\omega t}$  in Eq. (1), and find the expression of  $\tilde{H}$ , such that  $\tilde{u} = \tilde{H}\delta_0$ .  $\tilde{H}$  is commonly referred to as the *transfer function* because it directly relates the input  $\delta_0$  to the output  $\tilde{u}$ .
- (d) The transfer function is characterized by its magnitude  $H$  and its argument  $\beta$ , so that  $\tilde{H} = He^{i\beta}$ . Determine the analytical expression of  $H$  and  $\tan \beta$ .

### 4. Let's study the response of the car as a function of the driving frequency, and observe resonance phenomenon.

- (a) We define  $v = 1/\omega^2$  and write  $H(v) = 1/\sqrt{g(v)}$ . Identify the function  $g$  and write  $H' = dH/dv$  as a function of  $g$  and  $g'$ .
- (b) Show that depending on the value of  $Q$ ,  $H$  may reach a maximum value.
- (c) Plot  $H$  and  $\beta$  as a function of  $\omega$  for the different regimes and interpret the obtained graphs. Identify the resonance phenomenon on the graphs.
- (d) How are the amplitude  $A$  and phase shift  $\Phi$  of the oscillations of the chassis related to  $H$  and  $\beta$ ? Describe the behavior of the car for different forcing frequencies  $\omega$ .
- (e) As a car manufacturer, you can adjust the characteristics of the suspensions to tune their natural frequency  $\omega_0$  and quality factor  $Q$ . Assuming road bumps only "shake" the car within a restricted frequency range, how would you then design your suspension to isolate the car from vibrations?
- (f) In the movie "le salaire de la peur", Yves Montand drives a truck (with given  $\omega_0$  and  $Q$ ) loaded with nitroglycerin. He is driving on a wavy metal sheet that has a given spatial wavelength and amplitude. To avoid any explosion, he has to drive over the metal sheet with a speed either lower than  $5 \text{ km.h}^{-1}$  or higher than  $50 \text{ km.h}^{-1}$ . Qualitatively explain this using previous results.

## BONUS

### EXERCISE 1. TUNED MASS DAMPER (TMD)

When a structure is subjected to an oscillating loading at its natural frequency, its oscillations may become too important and lead to its ruin. In order to avoid such phenomenon a common practice is to use tuned mass dampers (TMD). The idea is to couple the structure to a harmonic oscillator with carefully chosen features so that it counteracts the resonance. A single degree of freedom representation of such a device is given in Figure 2a.

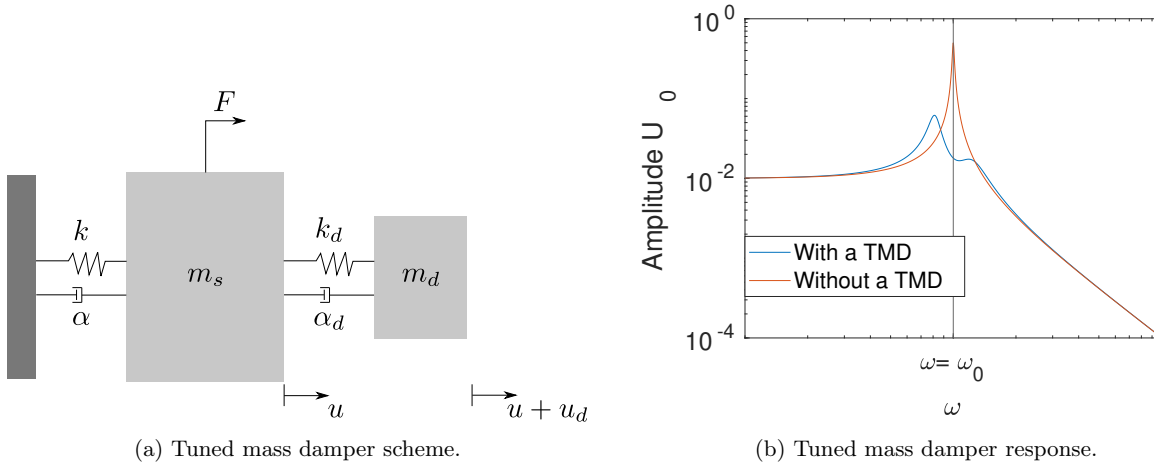


Figure 2: TMD principle.

- (a) Show that the equations of motion of the TMD mass and of the structure are coupled and read

$$\begin{cases} (1 + \bar{m})\ddot{u} + 2\xi\omega_0\dot{u} + \omega_0^2 u = \bar{F} - \bar{m}\ddot{u}_d \\ \ddot{u}_d + 2\xi_d\omega_d\dot{u}_d + \omega_d^2 u_d = -\ddot{u} \end{cases} \quad (2)$$

with  $\bar{m} = \frac{m_d}{m_s}$  and give the expression of  $\xi$ ,  $\xi_d$ ,  $\omega$ ,  $\omega_d$  and  $\bar{F}$ .

We now assume a harmonic form for the driving force  $F(t) = \tilde{F} \cos(\omega t)$ . We also consider the steady state of the forced oscillations.

- (b) Using complex representation, find the forced solution of the equation of motion of the structure when using a TMD.

A quick way to tune the TMD is to build it so that its natural frequency matches that of the structure. Doing so lead to frequency responses as the one plotted in Figure 2b.

- (c) Compare the amplitude of the oscillations of the structure with TMD at its natural frequency and the oscillation of the mass of the TMD at the same frequency.

- (d) Compare the amplitude of the oscillations of the structure with and without TMD at its natural frequency, what would the equivalent damping coefficient of a single damped harmonic oscillator need to be in order for its displacement magnitude to match the one of the structure using a TMD at the natural frequency? Discuss the results.

### EXERCISE 2. CAR STIFFNESS ESTIMATION

You and three of your friends weight 82 kg and are driving over a “washboard” dirt road with corrugations 4.0 m apart. You notice that the car bounces with maximum amplitude when your drive at 16 km/h. By how much does the car body rise on its suspension when you all get out of the car?

# OPhO Invitational: Fluid Dynamics

August 1, 2022

## 1 Introduction

Fluid dynamics is an important area in physics and engineering, and being able to simulate fluids properly is a challenging task that is applicable for rocketry, airplanes, cars, and even in designing pipes for air conditioning! The most popular technique is **computational fluid dynamics** (CFD), which numerically solves the Navier-Stokes equation. However, it is very difficult to set up and is very slow.

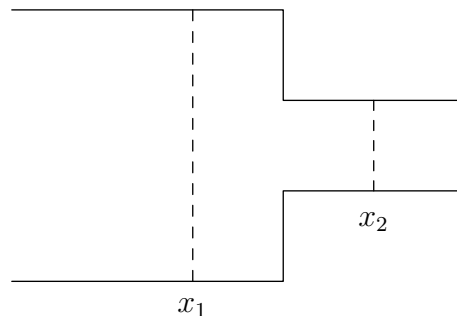
In this experiment, we will use a very simple simulation software that simulates fluids in two dimensions through the Lattice-Boltzmann algorithm (and is thus, *not* CFD), which you can find [here](#).

Please take time to read the description on the page. **Make sure to keep all settings as default.** By dragging your cursor around, you can create different types of barriers, and clicking start will run the simulation. The first dropdown menu tells us the number of pixels. While a higher resolution will lead to more accurate results, it will be more strenuous for the computer and harder to create boundaries. We recommend you to draw the barriers on a lower resolution but run the simulation on a higher resolution.

On the last row, we recommend checking **Flowlines** and **Sensor**. Enabling sensor will result in a cursor to appear on the screen that you can drag around, where the density  $\rho$ , horizontal speed  $u_x$ , and vertical speed  $u_y$  will all be shown. If for some reason the simulation is unstable, you may also check the **Data** button, which will record the data on the cursor as a function of time.

## 2 Mass Flow Conservation (15 pts)

We will first investigate if this implementation of the Lattice-Boltzmann algorithm correctly simulates the conservation of mass.



1. Create a tube that either opens up or closes down very abruptly such as in the diagram above. Pick two locations  $x_1$  and  $x_2$  such that one is in the wider section and the other is in the more narrow section. Compute the flow  $\frac{dm}{dt}$  past both these points. (10 pts)
2. Do the two mass flow rates agree with each other? If not, suggest a possible reason why. (5 pts)

Note:

- You are free to pick the shape and dimensions of the tube and it does not need to be perfect.
- Please report your figures with uncertainties and make clear how you estimated and/or obtained them! You do not need to show your computations.
- Provide a screenshot of the boundaries and the parameters.

### 3 Law of the Wall (25 pts)

The **Law of the Wall** provides an empirical relationship between the average velocity of a turbulent flow at some point and the distance to a wall (or fluid boundary). Namely,

$$u_x = \frac{u_T}{\kappa} \ln \left( \frac{\rho u_T}{\mu} y \right) + C,$$

where

- $u_x$  is the component of velocity parallel to the wall
- $u_T$  is the shear velocity defined by  $u_T = \sqrt{\tau_W / \rho}$ ,
- $\rho$  is the density
- $\mu$  is the viscosity (specifically, the dynamic viscosity)
- $\kappa, C$  are constants.

Note that  $\tau_w = \mu \frac{\partial u_x}{\partial y} \Big|_{y=0}$  is the wall shear stress.

Studies have shown that  $\kappa$  ranges from 0.35 to 0.42. Using a straight line to model a wall, answer the following questions:

1. Is the relationship between  $u_x$  and  $y$  logarithmic? (10 pts)
2. What are the values of  $\kappa$  and  $C$ ? (10 pts)
3. Does the value of  $\kappa$  agree with experiment? If it does not, suggest a possible reason why. (5 pts)

You are encouraged to add graphs.

Note:

- You are free to pick the length and direction of the boundary.
- Please report your figures with uncertainties and make clear how you estimated and/or obtained them! You do not need to show your computations.
- Provide a screenshot of the boundaries and the parameters.